we want to record data, specified as `runningTime` in minutes, and the interval in between sequential readings, specified as `interval` in milliseconds. We then step through the initialization process providing feedback through the Serial Monitor at every step. Once the SD library is set up, we need to open a file to write our data to, set our clock for the running time, and then we enter a `while` loop that, for as long as the duration lasts, will print the analog sensor data to both the Serial Monitor and to the file on the SD card at an interval determined at the beginning of the sketch. After the time has expired, the file is closed and the sketch effectively ends. Each time the sketch is restarted, through reset or cycling power, it will continue to amend data to the existing values already in the file. To start over, the file will need to be deleted first.

Listing 9-8. *SD Logger Source Code*

```
#include <SD.h>

File file;

const long runningTime = 5;  // time in minutes
const long msMin = 60000;    // milliseconds in 1 minute
const int interval = 2000;   // time in milliseconds

const int sensorMin = 30;
const int sensorMax = 1000;
const int sensor = A0;
const int SDcs = 4;

void setup() {
  Serial.begin(9600);
  Serial.print("Initializing SD card...");
  pinMode(10, OUTPUT);

  if (!SD.begin(SDcs)) {
    Serial.println("initialization failed!");
    return;
  }
  Serial.println("initialization done.");

  file = SD.open("data.txt", FILE_WRITE);

  if (file) {
    while (millis() < (runningTime * msMin)) {
      byte value = map(analogRead(sensor), sensorMin, sensorMax, 0, 255);

      Serial.print("Writing ");
      Serial.print(value, DEC);
      Serial.print(" to data.txt... ");
      file.write(value);
      Serial.println("done.");
      delay(interval);
    }
    file.close();
    Serial.println("File closed.");
```